AgenticTactileVLA: Contact-Guided Execution-Time Supervision for Generalizable Dexterous Manipulation without VLA Retraining

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***Abstract*— Vision-language-action policies may predict a transferable manipulation strategy yet fail to realize it reliably on the encountered object: objects compatible with the same grasp differ in geometry and compliance, and visual feedback degrades under closure occlusion. AgenticTactileVLA is presented as an execution-time supervisor that shifts part of object-specific adaptation from prediction to physical interaction. A fixed VLA provides the approach and hand targets; the supervisor decides whether to remain transparent, refine finger flexion, retain or release the corrected configuration, return control to the VLA for retry, or select a compliant hand-control regime. It uses finger-position and motor-effort feedback as proprioceptive contact evidence and requires neither tactile sensors nor VLA retraining. On a Unitree G1 with a BrainCo Revo2 hand, a randomized matched-block evaluation on five objects held out from VLA training yields 61.3% completion for the base VLA, 72.0% for unconditional close-to-stall control, and 84.0% for the supervisor under a shared budget; the gain is positive on every object and persists under moderate pose perturbations. Ablations show the gain is not explained by extended closure alone, and that selective triggering reduces correction episodes by 65.7% with no detected change in completion. A retention audit shows acceptance predicts retention in 88.9% of held-out cases, while compliant objects expose conservative false rejection. A thin-walled-cup study demonstrates contextual routing to compliant control, matching an always-compliant reference. These results suggest that contact-guided execution-time adaptation can improve the object-level generalization of a fixed VLA to held-out objects by adapting physical realization without object-specific retraining.**

I. Introduction

A central challenge in robot learning is to generalize manipulation policies to new objects while maintaining reliable physical execution. For vision-language-action (VLA) models, this is particularly important in contact-rich manipulation. Objects compatible with the same broad grasp strategy may differ in geometry and compliance, changing the finger contacts and interaction conditions required for successful execution. A VLA may therefore generate a suitable approach yet still leave fingers without sufficient support, produce an unsuitable terminal hand configuration, or over-compress a fragile object. Such failures are difficult to resolve visually during closure, when the hand increasingly occludes the contact region.

Many objects can be handled using a limited repertoire of broad grasp strategies [1], [2], while their physical realization may still require object-specific finger contacts and interaction effort. This motivates reusing a fixed policy while adapting these physical details during execution rather than through object-specific retraining.

We investigate a division of responsibility in which the VLA provides the transferable manipulation strategy, while an execution-time supervisor adapts its physical realization to the encountered object (Fig. 1). The supervisor can modify finger flexion and gate progression of the VLA-generated body reference, but does not replace the VLA approach policy or learn a residual action model. Prior work has explored tactile-conditioned VLA adaptation [3], [4] and execution monitoring and recovery [5], [6]. Our work lies at their intersection and asks: *Can execution-time physical adaptation improve the realization of a fixed VLA policy on objects outside its training set without object-specific retraining?*

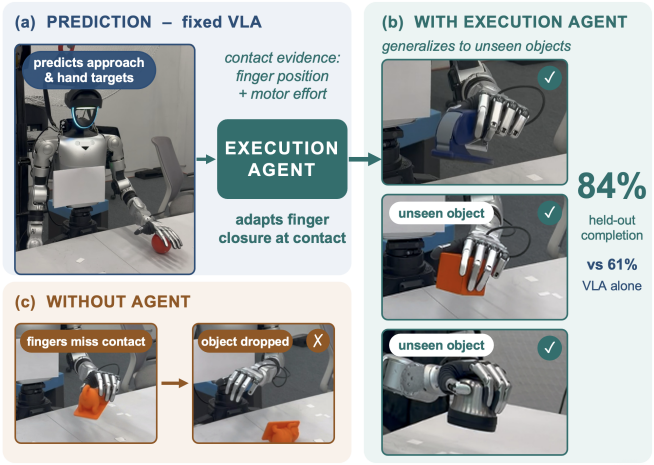


Fig. 1. From VLA prediction to execution-time physical adaptation. The fixed VLA provides the approach and initial hand configuration, while the execution agent adapts finger closure to the encountered object.

AgenticTactileVLA augments a fixed VLA with a stateful execution supervisor. At the terminal phase of nominal closure, it evaluates proprioceptive support evidence. If support is sufficient, execution remains transparent to the policy. Otherwise, body-reference progression is paused while whole-body stabilization remains active, and unsupported fingers are corrected independently. An accepted correction is retained; a rejected correction releases the hand override and returns control to the VLA for subsequent policy-generated retry.

A key principle is *selective intervention*: the supervisor acts only when the policy-proposed closure is not sufficiently realized physically. To separate this effect from simply closing farther or longer, we compare against unconditional close-to-stall control under the same execution budget and use component ablations to examine selectivity, effort-assisted stopping, and verdict-dependent release.

Geometric correction alone is insufficient when the interaction regime must also change. Fragile objects may deform

before proprioceptive resistance is large enough for reactive correction. AgenticTactileVLA therefore also selects between nominal position control and a predefined compliant mode before closure. Explicit handling cues can request compliant interaction, while a vision-language model (VLM) can select between the two modes from image and task context. The VLM selects only the interaction mode and does not generate controller parameters or material estimates.

We use the term *agentic* to denote this bounded, stateful supervision: deciding whether to remain transparent or intervene, adapting individual finger references, accepting or releasing the corrected grasp, returning control to the VLA for retry, and selecting the interaction regime. This distinguishes the layer from low-level reflexes by enabling high-level decision-making based on multimodal context. The framework uses finger position and motor-effort signals as device-specific proprioceptive evidence and requires neither additional tactile sensors nor object-specific VLA retraining.

The main contributions of this work are:

1) A contact-guided execution-time supervisor for a fixed VLA, combining selective intervention, bounded per-finger correction, grasp acceptance or release, and policy-mediated retry.
2) A proprioceptive execution mechanism using finger position and motor-effort feedback without additional tactile hardware, together with language- or vision-conditioned selection between nominal and compliant interaction modes.
3) Real-hardware evaluation on a Unitree G1 with a BrainCo Revo2 hand under matched execution budgets, including unconditional close-to-stall control, component ablations, pose perturbations, retention diagnostics, and fragile-object interaction tests.

The intended scope is adaptation of feasible policy-generated manipulation strategies, not recovery from arbitrary incompatible approach errors. Within this scope, we view execution-time physical adaptation as a step toward more scalable object-level generalization of fixed VLA policies, reducing the need for object-specific retraining while preserving the policy's reusable manipulation strategy.

II. Related Work

A. Contact Sensing from Motor Feedback

Motor feedback can reveal physical robot–object interaction without dedicated tactile sensors. Current as Touch [7], Transferring Contact, Not Just Motion [8], and Pseudo-Tactile [9] use actuator or gripper signals for contact-aware manipulation.

AgenticTactileVLA uses finger positions and motor-effort signals as device-specific evidence of interaction resistance rather than calibrated contact force. This evidence determines whether the nominal closure provides sufficient support and, when necessary, guides local finger correction and grasp acceptance or release.

B. Tactile Feedback for VLA Adaptation

Tactile feedback complements vision when physical interaction is difficult to infer from images alone. VLA-Touch [3] combines a frozen VLA with semantic tactile feedback and learned action refinement. TLA [10] and VTLA [11] incorporate tactile observations into action generation, while UniTacVLA [4] predicts contact changes to refine action chunks. RoboTacDex [12] provides multimodal data for visual–tactile dexterous humanoid manipulation.

In contrast, AgenticTactileVLA does not train a tactile-conditioned VLA or a learned residual policy. Adaptation is placed in an external, nonlearned execution layer while the VLA remains fixed. The supervisor stays transparent when the policy-generated closure already provides sufficient support and intervenes only when physical correction is required.

C. Execution Supervision and Interaction Control

Execution supervision connects predicted actions with observed consequences. PLanAR [5] verifies expected effects and replans after deviations, VoLo [6] combines monitoring with recovery, and VLAs Fail Differently [13] analyzes failure structure in motor-command sequences.

This work brings supervision to the closure-to-motion transition in dexterous manipulation. The supervisor determines whether the physical realization of the VLA action is sufficient, whether individual fingers should be corrected, whether the resulting grasp should be retained or released, and whether control should return to the VLA for another approach. A separate interaction-mode layer selects between nominal position control and current-limited compliant control before closure.

AgenticTactileVLA thus combines selective intervention, contact-guided finger adaptation, grasp acceptance or release, policy-mediated retry, and contextual interaction-mode selection within a single execution-time layer over a fixed VLA policy.

III. Method

The architecture separates policy prediction from physical execution. The fixed VLA generates body-motion tokens and hand targets, while the execution supervisor monitors how these commands are physically realized. When the policy-generated grasp is sufficiently realized, execution remains unchanged. Otherwise, the supervisor can pause body-reference progression, adapt individual fingers, or return control to the VLA for another approach. A separate interaction-mode layer selects nominal or compliant hand control before closure, while whole-body stabilization remains active throughout execution (Fig. 2).

A. System Overview and Execution Supervisor

a) Nominal VLA execution: GR00T-N1.6-3B [14], fine-tuned on teleoperated manipulation demonstrations, is deployed on a Unitree G1 with a BrainCo Revo2 hand and remains fixed during agent-assisted execution.

Let o_t denote the policy observation, including the ego-view image, robot state, and measured hand positions, and let

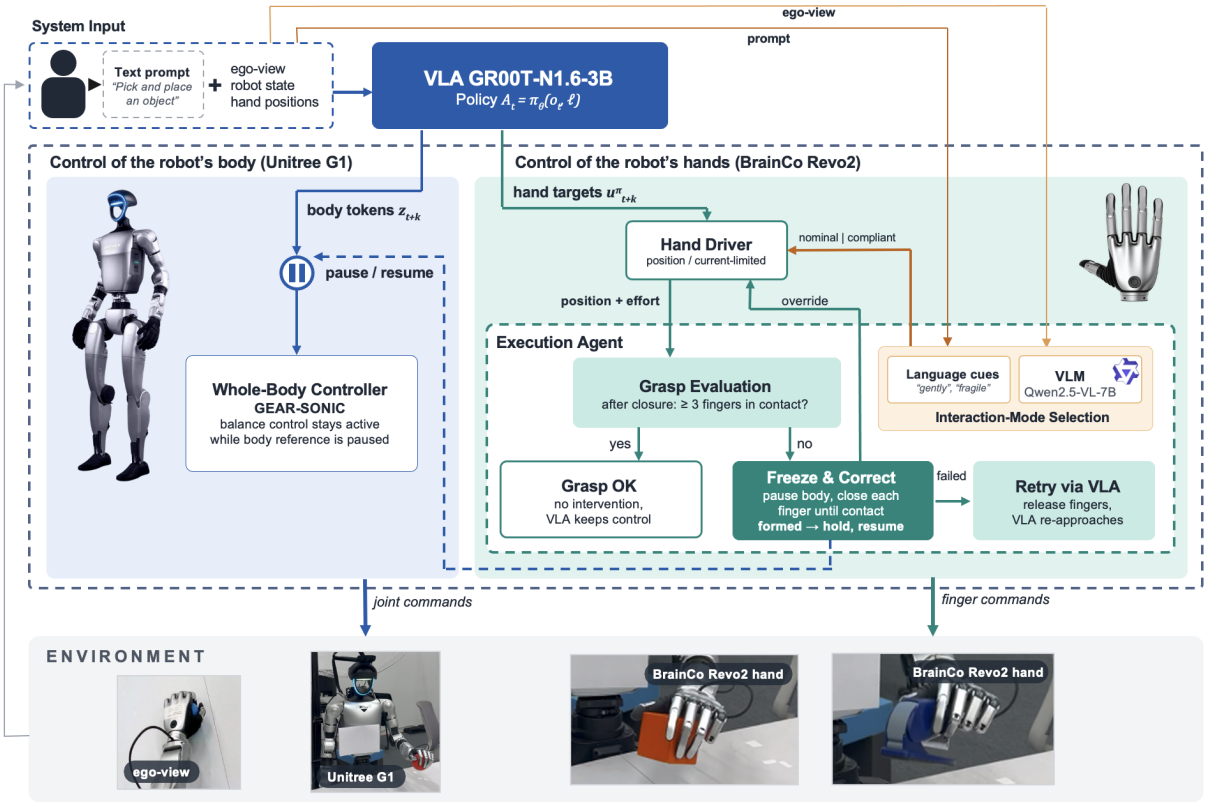


Fig. 2. System architecture of AgenticTactileVLA. The fixed VLA provides the nominal manipulation strategy, while the execution supervisor monitors its physical realization and intervenes only when adaptation is required.

ℓ denote the natural-language instruction. The VLA produces an action chunk

$$A_t = \pi_\theta(o_t, \ell) = \left(z_{t+k}, \mathbf{u}^{\pi}_{t+k}\right)_{k=0}^{H-1}, \tag{1}$$

where z_{t+k} is the body-motion token and $\mathbf{u}^{\pi}_{t+k} \in [0,1]^6$ contains the nominal position targets for the six hand channels.

GEAR-SONIC [15] realizes the body-motion tokens, while a dedicated driver executes thumb flexion (0), thumb opposition (1), index (2), middle (3), ring (4), and little finger (5). Thumb opposition preserves the policy-selected grasp arrangement; execution-time correction modifies only the flexion channels.

The hand publishes measured finger positions $\mathbf{q}_t \in [0,1]^6$ and a motor-effort proxy $\mathbf{e}_t$. These signals provide the proprioceptive feedback used by the supervisor. Motor effort is treated as a device-specific indicator of interaction resistance, not as a calibrated measurement of contact force.

b) Agentic execution supervisor: The execution agent wraps the fixed VLA as a stateful supervisory layer:

$$(m_t, \gamma_t, \mathbf{r}_t, b_t) = \mathcal{S}\left(A_t, \mathcal{H}_t, \ell, I_t, \sigma_t\right), \tag{2}$$

where $\mathcal{H}_t$ stores issued hand references, measured positions, and motor-effort feedback; I_t is the visual observation and σ_t the internal execution state.

The outputs define execution authority. The mode m_t selects the hand-control regime, γ_t selects nominal execution or intervention, $\mathbf{r}_t$ contains override finger references, and b_t gates progression of the body reference. The agent does not replace the VLA arm policy: body motion continues to originate from policy-generated tokens.

The supervisor operates at the boundary between *policy completion* and *physical grasp completion*. A closure is visible in the action chunk as progressive flexion-target motion. In the reference realization, the closure is armed when any flexion target advances by 0.015 over $0.25\,\mathrm{s}$ and reaches its terminal phase when every flexion target varies by less than 0.005 over $0.12\,\mathrm{s}$. This terminal phase defines the grasp-assessment point: the VLA has completed its commanded grasp, but the supervisor must still determine whether sufficient physical support has actually been established.

Position-based support evidence is accumulated from arming, while effort-relative evidence is introduced only after entering ADJUST. If support is sufficient, the supervisor remains transparent. Otherwise, body-reference progression is paused and the flexion targets are locally corrected. Missing evidence at this stage does not by itself prove absence of physical contact.

An accepted correction is retained while body-reference progression resumes. A rejected correction releases the hand override and returns execution authority to the VLA. At authority transitions, stale queued chunks are discarded and the next VLA chunk is generated from a fresh observation; whole-body stabilization holds the current body reference while the new chunk is pending. Thus, the corrected physical hand state can directly condition subsequent VLA execution.

TABLE I

EXECUTION-AGENT STATES. b_t GATES BODY-REFERENCE PROGRESSION; THE OVERRIDE COLUMN INDICATES WHETHER THE AGENT REPLACES THE NOMINAL HAND TARGET.

State	b_t	Override	Transition / exit condition
IDLE	0	no	upcoming closure in the action chunk
ARMED	0	no	terminal grasp assessment
ADJUST	1	yes	grasp verdict or timeout
HOLD	0	yes	policy reopens the hand

c) Control branches and execution timing: The supervisor has two physical execution branches. The position branch preserves the VLA grasp strategy and supports contact-guided finger correction. The compliant branch uses current-limited interaction control and can be selected before closure when language or visual context indicates that interaction effort should be constrained.

Finger references are issued at 50 Hz. When $b_t = 1$, only progression of the body reference is paused; whole-body stabilization remains active. Table I summarizes the supervisory states and authority transitions.

B. Selective Intervention

Selective intervention determines when physical execution should transfer from the VLA to the supervisor. It is evaluated only at the grasp-assessment point; before this point, missing contact is expected and does not trigger correction.

Let $\mathcal{F} = \{0, 2, 3, 4, 5\}$ denote the monitored flexion channels, and let $\hat{c}_i(t) \in \{0, 1\}$ denote the position-support evidence accumulated since arming for channel $i \in \mathcal{F}$. Let g_t indicate that the VLA has reached the terminal phase of its commanded grasp. Intervention is defined as:

$$\eta_t = g_t \, \mathbb{I}\left[\sum_{i \in \mathcal{F}} \hat{c}_i(t) < 3\right], \tag{3}$$

where $\eta_t = 1$ transfers hand execution from the nominal VLA command to the execution agent.

At least three support latches allow nominal execution to continue. Otherwise, the supervisor sets $b_t = 1$, overrides the flexion targets, and enters ADJUST. The three-support criterion is an operational acceptance rule, not a guarantee of grasp stability.

C. Proprioceptive Contact Estimation

Contact is estimated only from proprioceptive signals available from the hand. For each monitored flexion channel $i \in \mathcal{F}$, let $u_i(t)$ denote commanded position, $q_i(t) \in [0, 1]$ measured normalized position, and $e_i(t)$ device-specific motor effort. Thumb opposition remains at the policy target.

The main cue is the mismatch between commanded and measured finger motion. Over a history window W,

$$\Delta u_i(t) = u_i(t) - u_i(t - W), \tag{4}$$

$$\Delta q_i(t) = q_i(t) - q_i(t - W). \tag{5}$$

When the command continues to advance but measured motion is substantially smaller, the finger is interpreted as encountering resistance.

Let $E_i(t) \in \{0, 1\}$ denote detector eligibility. Eligibility requires a complete history window without reopening and at least δ_u commanded advance; a saturated target does not create additional evidence. Position-based stall is

$$s_i(t) = E_i(t) \, \mathbb{I}[\Delta u_i(t) \geq \delta_u \wedge \Delta q_i(t) < \rho_s \Delta u_i(t)], \tag{6}$$

where ρ_s bounds the fraction of commanded motion reproduced by the measured finger.

Motor effort provides complementary evidence during adjustment. At entry to ADJUST, the current feedback sample establishes a fixed per-channel baseline b_i, and the effort-duration accumulator is reset. Let $D_i(t)$ be the continuous duration for which

$$|e_i(t) - b_i| > \delta_e.$$

Sustained effort deviation is combined with incomplete tracking:

$$\begin{aligned} d_i(t) = E_i(t) \, \mathbb{I}\big[&D_i(t) \geq T_e \wedge \Delta u_i(t) \geq \delta_u \\ &\wedge \Delta q_i(t) < \rho_d \Delta u_i(t)\big], \end{aligned} \tag{7}$$

where T_e is the required effort-deviation duration and ρ_d the corresponding tracking-ratio threshold.

Support for finger i is latched when $s_i(t) \vee d_i(t)$ activates. Pre-assessment position latches are preserved, effort-assisted detections accumulate only in ADJUST, and reopening resets all latches. Missing activation can therefore indicate compliance or insufficient excitation rather than true absence of contact.

The deployed detector parameters are

$$W = 0.25\,\text{s}, \qquad \delta_u = 0.015, \qquad \rho_s = 0.3,$$

$$\rho_d = 0.7, \qquad \delta_e = 0.3, \qquad T_e = 0.06\,\text{s}.$$

a) Parameter selection: Detector thresholds were selected from 40 calibration closures on the five training objects (eight per object), separate from all evaluation trials. Calibration targeted stable finger stopping and suppression of short effort transients. Held-out objects were not used for tuning, and all thresholds, preload, timing parameters, and controller gains were frozen before evaluation.

Highly compliant objects remain challenging because deformation can preserve command tracking despite physical contact, reducing the command–response mismatch required for detection.

D. Object-Adaptive Finger Correction and Acceptance

Once intervention transfers execution to the supervisor, each unsupported finger advances independently toward its permitted travel limit. When support is detected, the reference is fixed at the measured position plus a normalized preload of 0.02, clipped to the travel limit; further advancement of that finger stops. Correction starts from measured positions and advances in increments of 0.004 every 20 ms.

Independent stopping transforms the VLA-proposed terminal hand configuration into object-dependent finger references: different fingers may stop at different depths and times, while the policy-selected thumb-opposition target preserves the overall grasp arrangement.

Reaching the travel limit without detector activation does not establish contact, because insufficient remaining motion may prevent the history and progress requirements of (6) from being satisfied.

Adjustment lasts at least 0.6 s and at most 4.0 s. After the minimum duration, it terminates when all monitored fingers have either latched support or exhausted their permitted travel, or when the timeout is reached. The final support count defines

$$v = \begin{cases} \textsc{Formed}, & \sum_{i \in \mathcal{F}} c_i \geq 3, \\ \textsc{Empty}, & \sum_{i \in \mathcal{F}} c_i = 0, \\ \textsc{Crooked}, & \text{otherwise.} \end{cases} \tag{8}$$

The verdict depends only on total support. CROOKED denotes insufficient support evidence for acceptance rather than geometric misalignment, while EMPTY denotes no detected support.

A FORMED verdict retains the corrected configuration and releases the body gate. An EMPTY or CROOKED verdict releases the hand override, allowing subsequent VLA actions to reopen the hand and generate another approach. Releasing the override does not itself command reopening.

To prevent repeated intervention on one closure, re-entry is blocked until the hand has reopened. A decrease of at least 0.05 in policy-commanded mean flexion relative to terminal assessment, sustained for 0.12 s, first releases a retained override. Rearming then requires both commanded and measured mean flexion to remain at least 0.05 below their terminal-assessment values for 0.12 s. Each closure therefore produces at most one correction episode, and all pauses and retries consume the same trial budget.

E. Language- and Vision-Conditioned Mode Selection

Interaction-mode selection chooses between nominal position control and compliant control. Explicit handling cues such as "gently", "carefully", or "fragile" activate compliant mode.

When no explicit cue is present, Qwen2.5-VL-7B-Instruct [16] selects between the two predefined modes from image and task context. The VLM runs asynchronously while the hand is open, with one decision cached per grasp episode. Language cues take priority; missing, late, or malformed VLM responses default to nominal control.

The VLM selects only the interaction mode and does not estimate material properties or generate controller parameters. Selection is performed before closure because a fragile object may deform before sufficient proprioceptive resistance can be detected.

To isolate mode selection in Section IV-F, the VLA receives the same task instruction in every condition. Handling cues are provided only to the interaction parser, while the VLM receives a fixed image-and-task prompt without those cues.

F. Compliant Control via Current-Limited Interaction

For compliant interaction, the agent activates a custom current-limited controller:

$$I_i^{\mathrm{cmd}}(t) = \mathrm{sat}_{[-I_{\max}, I_{\max}]} \left(K_p [q_i^d(t) - q_i(t)] - K_d \bar{\dot{q}}_i(t) \right), \tag{9}$$

where q_i^d and q_i are normalized hand-interface positions, $\bar{\dot{q}}_i$ is the exponentially smoothed finger velocity, and I_i^{cmd} is a device-level actuator command rather than calibrated motor current.

We use $K_p = 5.0$ and $K_d = 0.05$. These are controller-interface gains rather than physical stiffness or damping coefficients. The controller runs every 20 ms and bounds the actuator command by $I_{\max}$ while driving the fingers toward a predefined grasp configuration.

Raw command limits of 150 and 200 device units correspond to normalized controller-interface limits of $I_{\max} = 0.15$ and 0.20, respectively. These quantities do not represent calibrated motor current or fingertip force.

Compliant mode is activated before closure to prevent the nominal position-control path from applying an initial high-effort squeeze. The current implementation supports compliant closure and holding but not autonomous transport; fragile-object experiments therefore evaluate object integrity during closure and holding only.

IV. EXPERIMENTAL RESULTS

We evaluate AgenticTactileVLA through a sequence of controlled experiments. The evaluation first establishes the task-level benefit of execution-time supervision, then separates this effect from unconditional finger closure. We further examine sensitivity to initial-pose perturbations, the physical meaning and failure modes of proprioceptive verification, and contextual selection of the interaction regime for fragile objects.

A. Experimental Protocol

All experiments use the same fixed VLA checkpoint. The five *held-out* test objects were excluded from VLA training, detector calibration, and controller development.

A *trial* starts from an initialized scene and may contain multiple closures. A *retry* requires policy-commanded reopening followed by a new closure. Task completion is recorded independently of the supervisor.

The primary comparison uses randomized, interleaved matched blocks under a shared execution budget. Within each block, all conditions start from a common initial configuration. A trial terminates at successful placement, 60 s, or three arm approaches, whichever occurs first. No started trial is excluded.

Paired task-level comparisons use two-sided exact McNemar tests with Holm correction over all pairwise contrasts reported within each experiment.

B. Main Comparison

The primary experiment compares the base VLA, unconditional close-to-stall control, and the full position-branch supervisor over 75 matched blocks (15 per held-out object).

The close-to-stall baseline isolates a simpler explanation for improvement: the robot may benefit merely from allowing the fingers to close farther. At each terminal grasp, this baseline pauses body-reference progression and advances finger flexion until the position-stall condition in (6), the travel limit, or the shared 4.0 s timeout. Unlike the supervisor, it acts at every closure and has no contact-count verdict or verdict-dependent release.

TABLE II

FIXED-BUDGET TASK COMPLETION UNDER RANDOMIZED, INTERLEAVED CONDITIONS. ALL CONDITIONS USE 15 TRIALS PER HELD-OUT OBJECT; INTERVALS ARE 95% WILSON.

Object	VLA	VLA + stall	Supervisor
Cube	9/15	11/15	13/15
Fish	9/15	11/15	13/15
Soy sauce bottle	11/15	12/15	14/15
Thermos lid	7/15	8/15	10/15
Hand cream tube	10/15	12/15	13/15
Pooled	46/75	54/75	63/75
Rate	61.3%	72.0%	84.0%
95% CI (%)	50.0–71.5	61.0–80.9	74.1–90.6

Completion increases from 61.3% for the base VLA and 72.0% for unconditional close-to-stall control to 84.0% with the supervisor. In the matched blocks, the supervisor succeeds where the base VLA fails in 20 cases, versus three in the opposite direction, yielding $p = 0.001$ after Holm correction. Against unconditional closure, the corresponding discordance is 11/2 ($p = 0.045$, Holm-corrected) (Table II).

Unconditional closure alone exceeds the base VLA in 12/4 discordant blocks ($p = 0.077$), recovering roughly half of the supervisor's gain but not significantly outperforming the base policy. This indicates that additional closure contributes to performance, but does not fully account for the supervisor's gain.

The improvement is also consistent across all five objects: the supervisor gains 3–4 successful trials over the base VLA and 1–2 over unconditional closure for every object. Thus, the primary result is not driven by one particularly favorable test object.

The supervisor intervenes in 27/75 trials (36%), producing 36 correction episodes, whereas the unconditional baseline extends finger closure at every terminal grasp, resulting in 101 extension episodes across the same 75 trials. This motivates the component ablation below, which separates structured supervisory decisions from simply extending every closure.

C. Component Ablation

To determine which components of the execution supervisor drive the observed improvement, we conduct a component ablation over 100 matched blocks. The analysis examines whether the gain arises simply from allowing the fingers to close farther, whether selective intervention reduces unnecessary corrections, and whether effort evidence and verdict-dependent release provide additional benefit.

We evaluate five variants. Here, *U* denotes unconditional intervention at every closure, while *S* denotes selective intervention only when the supervisor detects insufficient support. *U-stall* simply extends every closure until position-based stall and serves as the additional-closure baseline. *U-full* applies the complete contact detector and grasp-verdict logic at every closure, testing whether structured supervision provides benefit even without selectivity. *S-position* retains selective intervention but removes motor-effort evidence. *S-retain* retains selective intervention but disables verdict-dependent release and policy-mediated retry. *S-full* is the complete selective supervisor (Table III).

TABLE III

COMPONENT-CONTROLLED ABLATION OVER 100 MATCHED BLOCKS. CORRECTIONS COUNT INTERVENTION EPISODES.

Variant	Completion	Rate	Corrections
U-stall	72/100	72%	135
U-full	84/100	84%	134
S-position	79/100	79%	49
S-retain	77/100	77%	47
S-full	86/100	86%	46

The full supervisor reaches 86% completion versus 72% for position-only stall control (Holm-corrected $p = 0.047$). The always-on variants show the same direction of effect: U-full reaches 84% compared with 72% for U-stall, although this contrast is not statistically resolved after correction ($p = 0.091$). Together, these results indicate that the gain is not explained by extended finger closure alone.

Comparing U-full with S-full isolates the role of selectivity. Their completion rates are similar (84% and 86%), but selective supervision reduces the number of correction episodes from 134 to 46, a 65.7% reduction. No completion difference is detected between the two variants ($p = 0.804$; paired 95% CI -5.8 to $+9.8$ points). Thus, selectivity mainly reduces unnecessary intervention while preserving the observed task-level performance.

Finally, S-position and S-retain test the contribution of individual supervisory mechanisms. Removing effort evidence reduces completion to 79%, while disabling verdict-dependent release reduces it to 77%. Neither individual contrast is statistically resolved after correction ($p = 0.378$ and 0.235, respectively). The ablation therefore supports the complete supervisory design over simple position-only closure, while the independent contribution of these two components remains unresolved.

D. Pose-Perturbation Evaluation

The previous experiments use matched initial configurations. We therefore test whether the supervisor's benefit disappears when the initial object pose is perturbed. The same five objects are evaluated under nominal poses, horizontal translations within 20 mm, and yaw changes within 15°, with 50 matched trials per regime and condition.

Performance decreases under perturbation for all methods, but the supervisor remains the strongest condition in every regime. Its gain over the base VLA is 22, 28, and 28

percentage points for nominal, translated, and rotated configurations, respectively. Pooled over regimes, the supervisor exceeds unconditional closure in 26/6 discordant blocks ($p < 0.001$); the individual per-regime contrasts at 50 trials are underpowered (Table IV).

TABLE IV
TASK COMPLETION UNDER INITIAL-POSE PERTURBATIONS.

Regime	VLA	VLA + stall	Supervisor
Nominal	32/50	37/50	43/50
Translation	25/50	32/50	39/50
Yaw	23/50	30/50	37/50
Pooled	80/150	99/150	119/150

The important observation is therefore not pose invariance, but persistence of the execution-time benefit within the tested perturbation range. The supervisor still depends on a feasible policy-generated approach and is not designed to recover from arbitrary approach errors.

E. Proprioceptive Verification and Failure Modes

The previous results show that the supervisor improves task completion, but they do not establish whether its internal support decisions correspond to a meaningful physical state. We therefore examine the contact-support mechanism separately from task success.

Across 70 diagnostic trials, the supervisor produces 85 correction events: 46 FORMED verdicts, 26 CROOKED verdicts, and 13 EMPTY verdicts. These internal decisions are not treated as contact-detection accuracy because independent per-finger contact labels are unavailable.

Contact patterns vary by object (Fig. 3). In particular, highly compliant objects expose a characteristic limitation of proprioceptive contact inference.

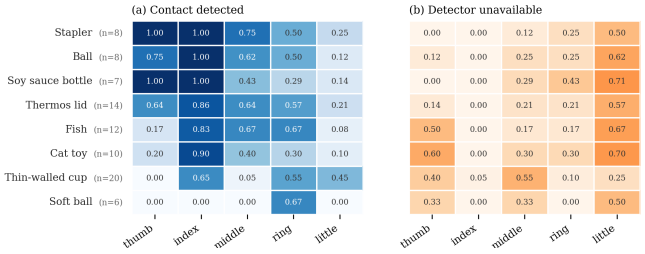


Fig. 3. Diagnostic contact detection (a) and detector ineligibility due to insufficient remaining finger travel (b).

a) Compliance-related false rejection: The stapler produces a FORMED verdict in 8/8 correction events and the ball in 5/8, whereas the soft ball produces none in six events. Video confirms that the soft ball is physically enclosed in all six cases. Its deformation allows the fingers to continue following their commanded motion, reducing the command–response mismatch used as contact evidence. Four events are therefore classified as CROOKED and two as EMPTY.

Despite these false rejections, five of the six trials eventually complete: three retain the object through placement and two succeed after a policy-generated retry. This shows that incorrect internal verdicts do not guarantee task failure, but highlights a clear limitation of motion-based proprioceptive inference.

TABLE V
SUPPORT VERDICT VERSUS OBJECT RETENTION DURING THE DIAGNOSTIC LIFT.

Support predicate	Retained	Not retained	Total
At least three	70	6	76
Below three	18	26	44
Total	88	32	120

b) External retention audit: To determine whether the support predicate nevertheless predicts a useful physical outcome, we perform 120 independent standardized closures (Table V). The grasp configuration is retained for a 5 cm diagnostic lift followed by a 2 s hold, providing an external retention label independent of the supervisor's verdict.

Accepted grasps retain the object in 70/76 cases (92.1%), showing that a positive support verdict is strongly associated with physical retention. Rejection is less reliable: 18/44 rejected configurations would still retain the object, corresponding to an NPV of 59.1%.

This asymmetry becomes stronger on held-out objects. There, retention given acceptance is 88.9%, while 25.6% of retainable configurations are falsely rejected. The support rule is therefore useful but conservative: acceptance provides strong evidence of a physically useful grasp, whereas rejection is less reliable, particularly for compliant or previously unseen objects.

F. Contextual Interaction-Mode Selection

The previous experiments adapt grasp realization during or after contact. Fragile objects pose a different challenge: damage may occur before sufficient resistance is observed. We therefore test whether the agent can select an alternative interaction regime before closure.

A thin-walled cup is evaluated under position control, instruction-based routing, VLM-based routing, combined instruction–VLM routing, and always-compliant control (Table VI).

TABLE VI
THIN-WALLED-CUP INTEGRITY AND COMPLIANT-MODE ACTIVATION.

Mode selected by	Trials	Intact	Damaged	Compliant active
No selection	5	0	5	0/5
Instruction cue	5	5	0	5/5
VLM from image	5	4	1	4/5
Instruction + VLM	5	5	0	5/5
Always compliant	5	5	0	5/5

Nominal position control damages all five cups, whereas instruction-based and combined routing preserve 5/5. VLM-only routing preserves 4/5; in the remaining trial, the correct compliant mode is selected after closure begins (Fig. 4(a)), showing that correct semantic selection must also arrive in time to affect the interaction.

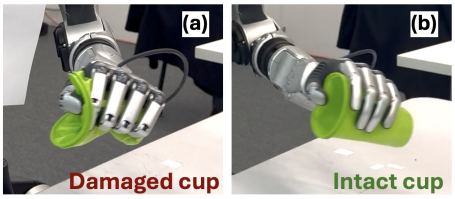


Fig. 4. Representative thin-walled-cup outcomes. (a) Damage under nominal position control. (b) Preserved integrity under compliant control.

Always-compliant control also preserves 5/5 cups, so the experiment demonstrates successful contextual routing rather than superiority over always-compliant execution (Fig. 4(b)). Since the cup is a training object, this does not establish held-out fragile-object generalization.

On three additional rigid objects, the intended nominal mode is selected in 14/15 trials. This confirms nonconstant routing between the two predefined interaction regimes, but not a general material classifier.

V. Discussion and Limitations

The experiments show that a fixed VLA benefits from an execution-time layer that adapts the physical realization of its predicted strategy. In the primary evaluation, the supervisor improves completion over both the fixed VLA and unconditional close-to-stall control, with gains on every held-out object. Additional finger closure explains only part of the improvement, while the ablation supports structured supervision over position-only stall control.

Selective intervention reduces correction episodes by 65.7% relative to the always-on controller without detected completion loss, and the benefit persists under moderate pose variation.

The retention audit shows that accepted configurations retain the object in 70/76 cases (92.1% pooled; 88.9% on held-out objects), while 18/44 rejected configurations would still retain it. The support predicate is therefore useful but conservative, especially for highly compliant objects where deformation weakens motion-based contact evidence. False rejections can still be recovered through policy-mediated retry.

Execution-time adaptation also extends to interaction-mode selection before contact. Instruction-based and combined routing preserve 5/5 fragile cups, while VLM-only routing preserves 4/5 due to late compliant-mode selection in one trial. On rigid objects, the intended nominal mode is selected in 14/15 trials.

Limitations remain. The ablation does not isolate the contributions of effort evidence and verdict-dependent release, and the lack of a detected difference between selective and always-on supervision does not imply strict equivalence. The supervisor relies on proprioceptive signals rather than calibrated force and cannot recover from arbitrary approach errors. The fragile-object study uses a training object, and compliant control is evaluated only for static grasping. Validation requires unseen objects and diverse embodiments.

Overall, the results support execution-time physical adaptation as a complement to a reusable VLA strategy while keeping the VLA fixed.

VI. Conclusion

AgenticTactileVLA introduces contact-guided execution-time supervision as an external adaptation layer for a fixed VLA. Rather than retraining for each object, the framework preserves the predicted strategy while adapting its physical realization through selective intervention, per-finger correction, grasp acceptance or policy-mediated retry, and interaction-mode selection.

On five held-out objects under matched execution budgets, the supervisor achieves 84.0% task completion versus 61.3% for the base VLA and 72.0% for unconditional close-to-stall control, with gains on every tested object. The benefit persists under moderate pose perturbations, selective supervision reduces intervention burden, and accepted configurations usually provide physically useful support. Fragile-object experiments further extend adaptation to interaction-mode selection before contact.

These results support execution-time physical adaptation as a route toward more scalable object-level generalization of fixed VLA policies. A reusable manipulation strategy can be complemented by an execution agent that adapts its physical realization without object-specific retraining or additional tactile hardware.

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